

Control biclustering structure and assay-block polarization

We applied biclustering to the normalized control-count matrices (C1, C2; Steps 14–15) using the biclustering wrapper (Step 18) to jointly partition genomic regions (rows) and assay/pair measurements (columns). The resulting heatmap (Figure 1) shows a pronounced block structure, resolving coherent row (region) and column (assay/pair) modules consistent with the region- and pair-cluster tables. Across columns, assay/pair measurements segregate into left-to-right blocks: signals are generally lower in the left and middle blocks and highest in the rightmost block. Across rows, distinct horizontal modules emerge: lower-third region modules exhibit strong enrichment confined to the rightmost assay block with concurrent depletion in the first two blocks; mid-level modules display a graded left-to-right increase; and upper modules remain broadly weak with only occasional narrow enriched stripes.

This polarized architecture indicates that a subset of assay/pair columns consistently captures control-enriched signal, whereas others contribute predominantly baseline or depleted profiles, and that genomic regions segregate into modules with shared assay-response patterns. Together, the column modules define assay/pair classes for block-wise contrasts and interpretation, while the row modules provide compact genomic units that summarize regulatory behavior. This modular backbone will anchor downstream characterization by enabling: (i) block-aware differential summaries and effect-size estimation across assay/pair classes; (ii) prioritization of right-block-enriched region modules for targeted analyses; and (iii) systematic annotation and enrichment testing of region modules to map them onto putative regulatory states. In aggregate, biclustering

furnishes a low-dimensional, interpretable scaffold on which subsequent regulatory inferences can be robustly organized.

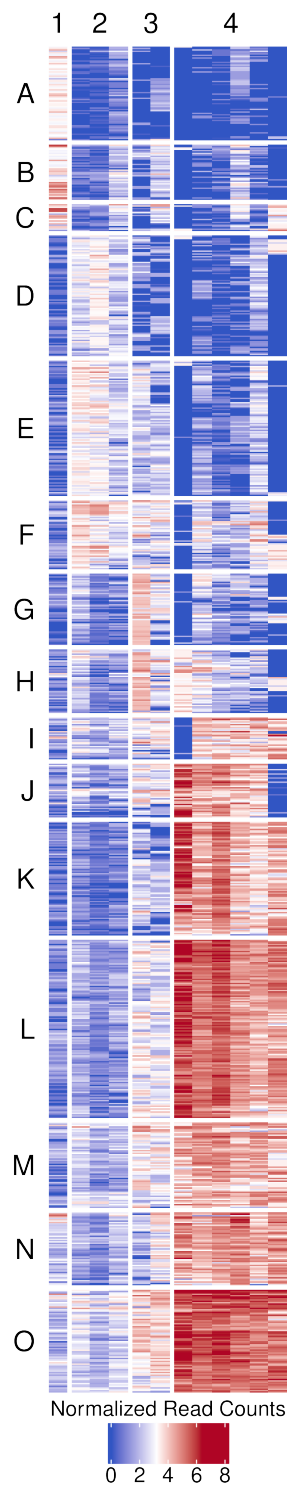


Figure 1. Biclustered control heatmap highlights right-block enrichment. *Normalized counts for C1–C2 are shown for biclustered regions (rows) and assay-pair modules (columns). Coherent blocks indicate that many region modules are depleted in the first two column modules and enriched in the rightmost module, while a subset remains uniformly low across columns.*

Regulatory characterization of control modules via TFBS, cCRE states, and genic context

To resolve regulatory programs and promoter–enhancer biases across control-region modules, we integrated TFBS enrichment patterns with cCRE states and genic context. The TFBS heatmap (Figure 2) separates region modules into a majority with broadly elevated motif/TFBS signals and a minority with attenuated signal. In parallel, cCRE composition (Figure 3) and gene-structure overlap (Figure 4) form a coordinated gradient from distal, intron- and intergenic-dominated modules to promoter-centric modules.

Promoter-like modules. Modules I–O are enriched for promoter-like signatures (PLS) with only minor enhancer-like fractions (Figure 3) and are predominantly assigned to promoter genic context with modest 5' UTR and exon overlap (Figure 4). These modules coincide with consistently elevated TFBS enrichment across many motifs (Figure 2), indicative of coherent promoter programs. Together, these features support a promoter-proximal regulatory architecture dominated by broadly active TFBS profiles.

Enhancer-dominated modules. Modules D–H show large distal enhancer-like (dELS) components with substantial proximal enhancer-like (pELS) contributions and minimal other cCRE categories (Figure 3). Their genic context is correspondingly intron- and intergenic-rich with relatively low promoter overlap (Figure 4), consistent with distal regulation. Within this set, Figure 2 reveals vertical bands of heightened motif enrichment confined to specific region modules, suggesting module-specific TF programs typical of enhancers. Module G is notable for its comparatively lower promoter overlap (Figure 4) and an extended pELS segment relative to dELS (Figure 3), indicating a bias toward promoter-proximal enhancers relative to neighbors D–E, which emphasize dELS.

Mixed/CA-rich modules. Modules A–C contain the largest other fractions alongside CA–H3K4me3, CA–TF, and TF categories (Figure 3), and show moderate intron/exon and minimal promoter assignment (Figure 4). These modules align with the weakest TFBS signals in Figure 2 (including the leftmost depleted group and a neighboring weaker group), suggesting either heterogeneous or less canonical regulatory content relative to the promoter- and enhancer-enriched modules.

Collectively, the concordance of broad TFBS enrichment with PLS- and promoter-heavy composition (modules I–O), and of motif-selective TFBS bands with dELS/pELS-

dominated, distal genic contexts (modules D–H), delineates promoter-centric versus enhancer-centric regulatory programs across the control-region modules.

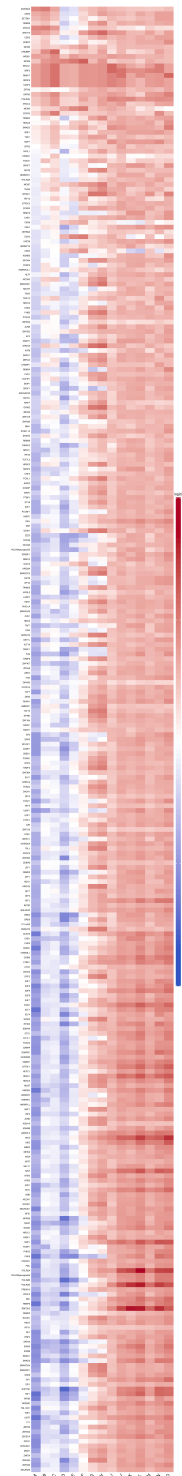


Figure 2. Divergent TFBS enrichment across C-group region clusters. *Transcription factor motif enrichment is summarized across region clusters from the C-group biclustering. Most clusters display broadly positive enrichment, whereas one leftmost cluster shows widespread depletion and an adjacent cluster is relatively weak, indicating distinct regulatory modules among control regions.*

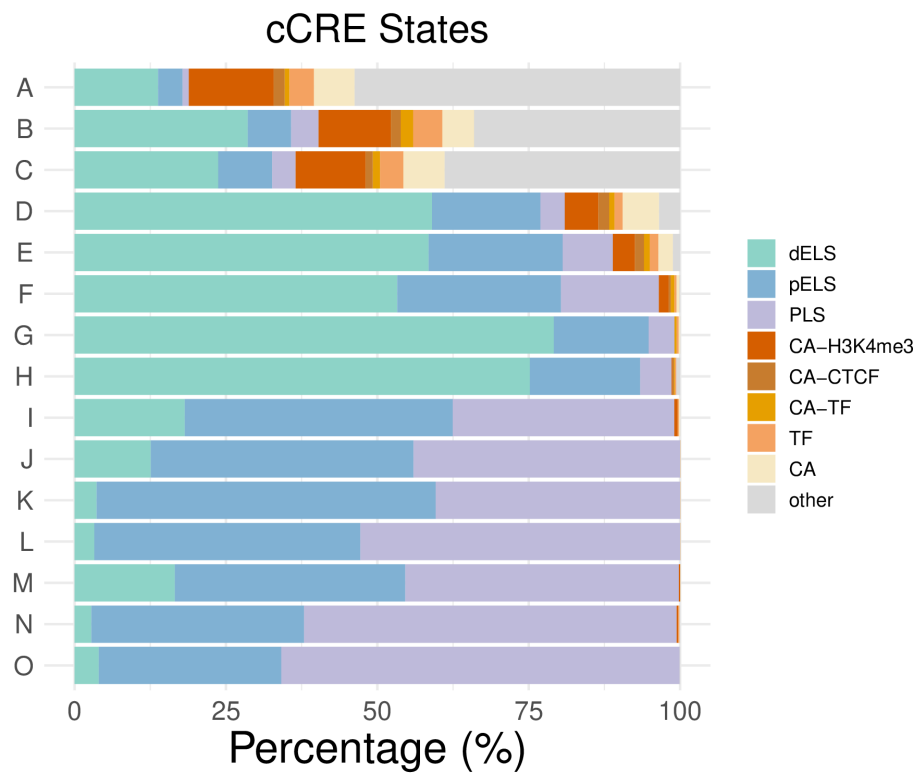


Figure 3. cCRE state composition across C-group bicluster modules. Proportions of cCRE categories are shown for C-group modules A–O. Promoter-like states (PLS) predominate in modules I–O, enhancer-like states (dELS/pELS) dominate D–H, and modules A–C retain larger ‘other’ and CA-related segments, including CA–H3K4me3 and CA–TF.

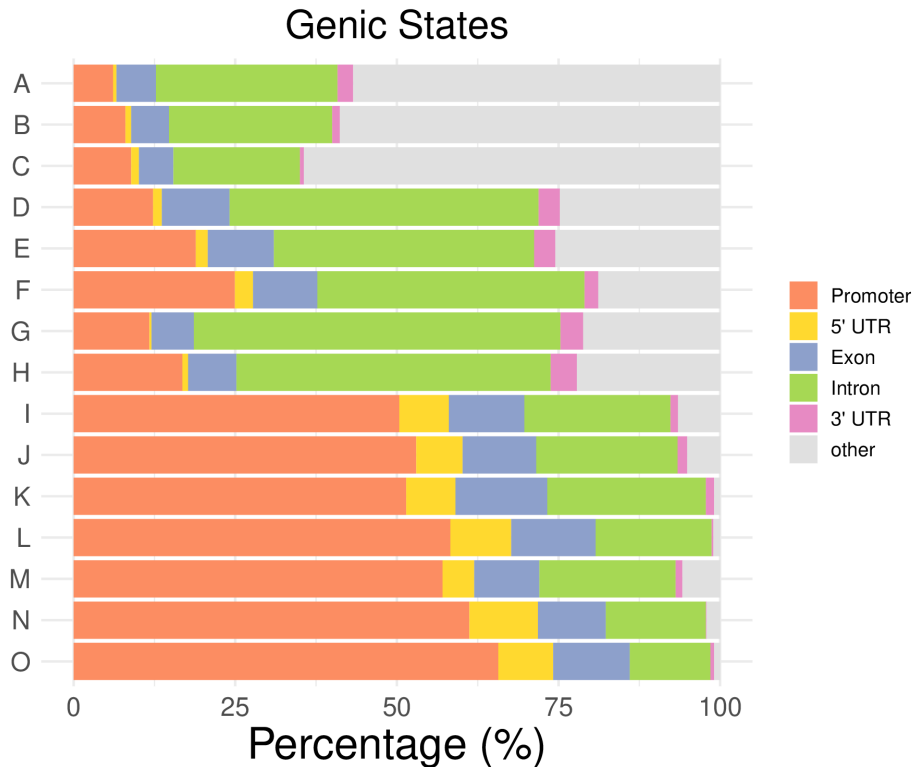


Figure 4. Genic annotations for C-group biclustered regions. Proportional distribution of Promoter, 5' UTR, Exon, Intron, 3' UTR, and other categories across C-group modules A–O. Later modules (I–O) are promoter-enriched with minimal other, whereas early modules (A–C) have higher other; mid modules (D–H) show prominent intronic contributions.

Differential accessibility by bicluster and pathway enrichment of treatment effects

Differential accessibility stratified by column clusters revealed that treatment-induced changes were highly uneven across the biclustered space (Figure 5). Clusters 2 and 4 accounted for virtually all significant differences, whereas clusters 1 and 3 showed no detectable effects. Across responsive clusters, there was a clear bias toward increased accessibility. Cluster 2 contained the highest number of differential regions in both directions, with a modest excess of up-regulated sites. Cluster 4 also exhibited both gains and losses but was dominated by up-regulated regions, indicating preferential chromatin opening under treatment in this subset.

Linking differential regions to putative target genes showed that pathway enrichment was strongly polarized to cluster 4 (Figure 6). Regions from cluster 4 were enriched for Hallmark Estrogen Response Early and Estrogen Response Late, alongside immune/inflammatory signaling signatures including IL2–STAT5 Signaling and TNFA Signaling via NF-κB. In contrast, cluster 2 showed broadly weak or absent enrichment, and most other pathways (e.g., Hypoxia, Glycolysis, KRAS Signaling, Interferon Responses, E2F Targets)

displayed minimal signal across both clusters. These results indicate that the predominant, up-regulated accessibility changes map to regulatory elements of estrogen-responsive programs and immune signaling axes concentrated in cluster 4, consistent with expected

engagement of ER-driven transcriptional circuitry under estrogenic or chromatin-modulating contexts (Johnson Thomas et al., 2026).

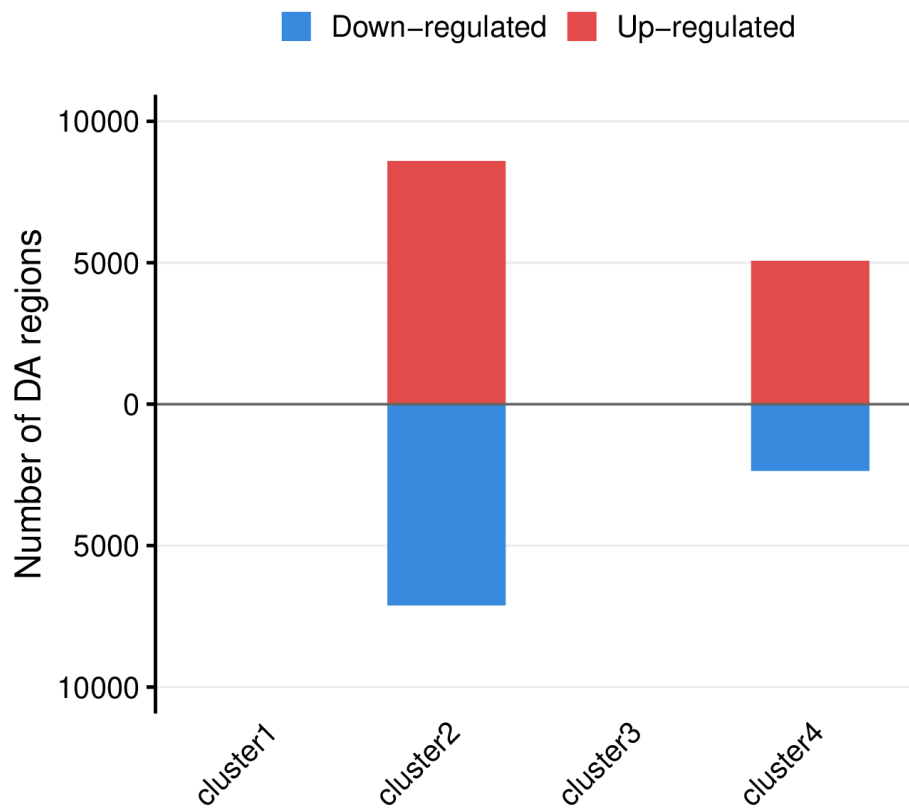


Figure 5. Directional DA region counts by bicluster across conditions. *Counts of up- and down-regulated DA regions are summarized per column cluster for the treatment versus control comparison. Changes are concentrated in clusters 2 and 4 with more up- than down-regulation, while clusters 1 and 3 show no detected differences.*

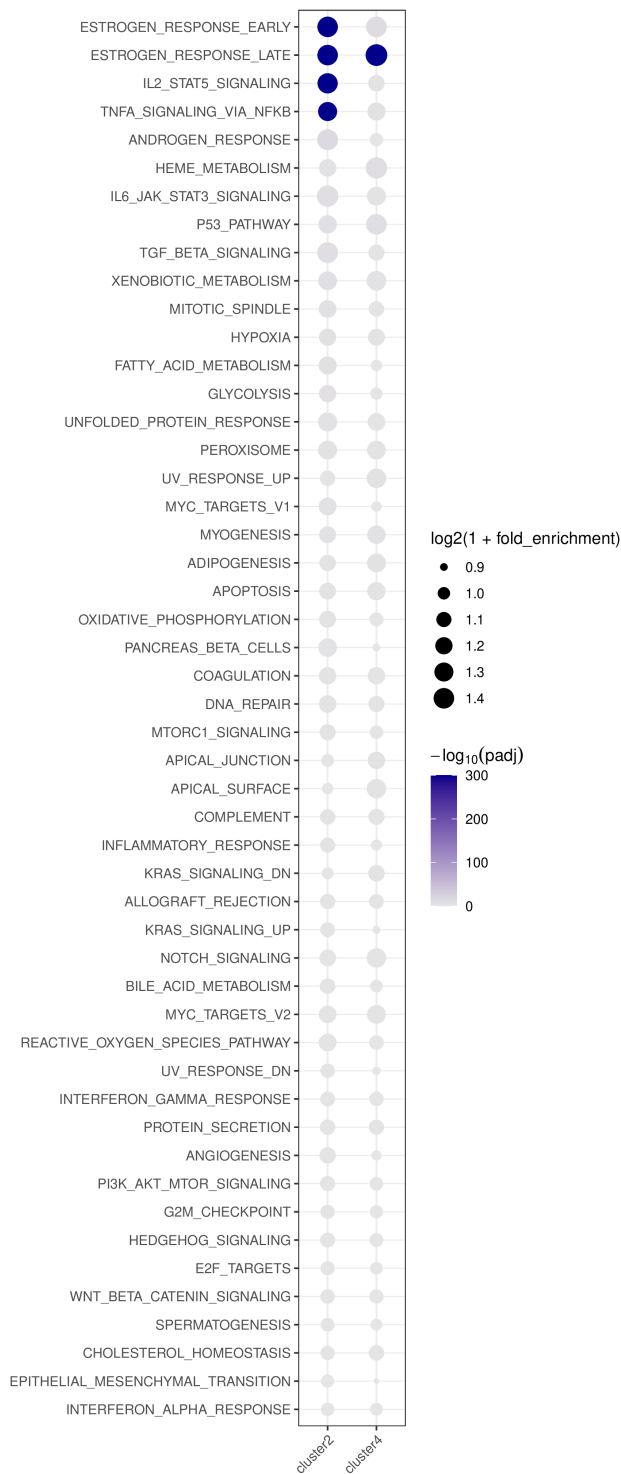


Figure 6. Pathway enrichment concentrates in cluster4 relative to cluster2. Bubble plot of enrichment for genes linked to T vs C differential regions. Among the two clusters, cluster4 shows enrichment for *ESTROGEN_RESPONSE_EARLY*, *ESTROGEN_RESPONSE_LATE*, *IL2_STAT5_SIGNALING*, and *TNFA_SIGNALING_VIA_NFKB*, whereas cluster2 shows little detectable enrichment across terms.

References

1. Liu, Y., Song, Q., & Chen, M. (2026). From linear methylation to spatial metabolic control: Rethinking DNMT1 targeting in MASLD: Letter to the editor on â DNMT1 facilitates the

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2. Johnson Thomas, A. L., Karakyriakou, B., Muskus, P. C., Roberts, A. M., Rendleman, J., Warren, A., Lu, H., Birmingham, K. A., Friedman, M. K., Goen, A. E., Yang, H., Wilkins, O. M., Kolling, F. W., Demidenko, E., Jorns, J. M., Christensen, B. C., & Miller, T. W. (2026). Pharmacological chromatin remodeling enhances response to estrogen therapy in ER⁺ breast cancer. Molecular Oncology. Portico. <https://doi.org/10.1002/1878-0261.70307>